

Exporter

Exporter [object]

Use the object *Exporter* for providing and exporting services. It represents a group of people, whose individual members you cannot distinguish and whom you cannot address as individuals.

Image



Description

The *Exporter* works together with the **Broker** and the *Importers* (see **Tab Importer** and **Sub-tab Failure**) of the *Station*, the *ParallelStation*, the *AssemblyStation*, and the *DismantleStation*. The *Exporter* offers services and provides them for *Importers*.

A single *Broker* manages the *Exporter* and assigns it to an *Importer*. After the *Exporter* has finished providing its service, it registers as being available with its *Broker* which will assign it to other *Importers* as soon as its services are required.

Note

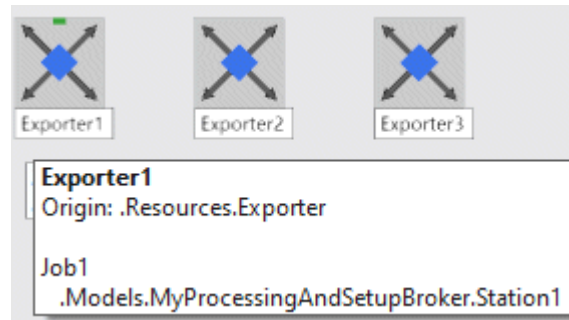
If the *Importer* requests several services at the same time, all of these services have to be available at the same time, before the *Broker* assigns them to the station.

Think of the *Exporter* as a group of people, whose individual members you cannot distinguish and whom you cannot address as individuals. To distinguish individual staff members, you have to model them with individual *Exporters* or **Workers**. An *Exporter* with a **capacity** greater than 1 can simultaneously provide services at several *Importers*. You can use the *Exporter* if transit times for traveled distances are not important for the simulation.

Note

An *Exporter* may provide services for several *Importers* at the same time.

To show a **tooltip** with information about the *Exporter*, hover with the mouse over it.

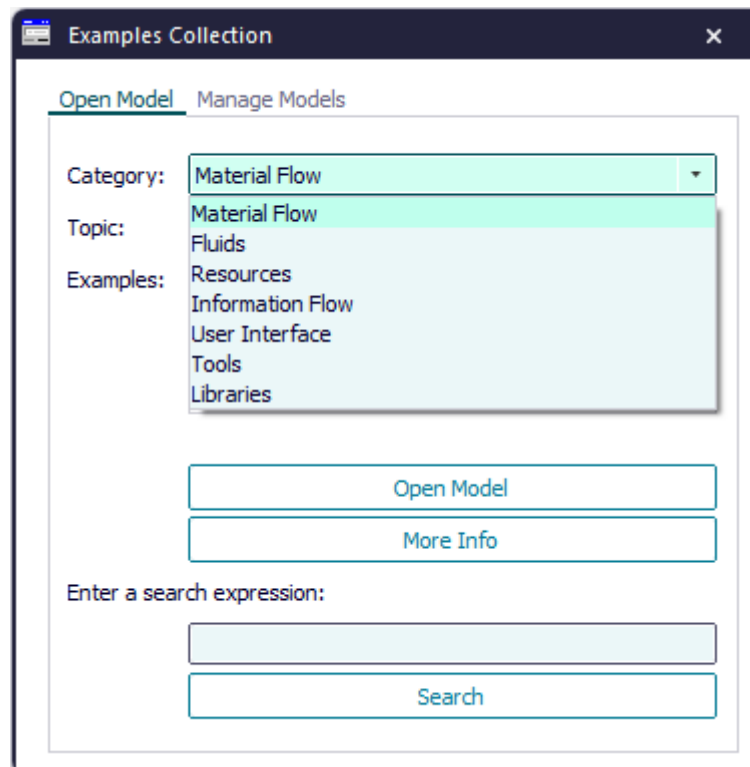


To change the length of the graphic and the anchor points of the *Exporter*, click [Show Manipulators](#)  on the **Edit** ribbon tab or press **M** on the keyboard.

Add the Object to the Simulation Model

To add the object *Exporter* to your simulation model, click  [Manage Class Library](#) > **Basic Objects** > **Resources** > **Exporter** on the **Home** ribbon tab.

Compare the sample models: Click the **Window** ribbon tab, click **Start Page** > **Getting Started** > **Example Models** > **Small Examples**. Then, select the respective **Category**, the **Topic**, and the **Example** in the dialog **Examples Collection**, and click **Open Model**.



See also

[Model Workers and the Jobs They Do](#)

[Model Workers with Importer, Broker, and Exporter](#)

Dialog Box of the Exporter

Dialog Box of the Exporter

Double-click the icon of the *Exporter* to open its dialog box.

Edit Simulation Properties

In the dialog box you can change the **simulation properties** of the object. The shared properties are described under [Dialog Items of the Objects](#).

Edit Animation Properties

To edit the **3D properties** of the object in the dialog box [Edit 3D Properties](#):

- Click the button  **Edit 3D Properties** in the lower left corner of the simulation properties dialog box.
- Select the object in the model and press the **spacebar**.

To manipulate the graphic of the object, click [Show Manipulators](#) on the **Edit** ribbon tab or press **M** on the keyboard.

See also

[Failed \[check box\] - material flow objects](#)

[Paused/Planned/Unplanned \[material flow objects\]](#)

Tab Attributes

Tab Attributes

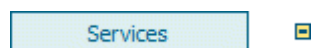
The tab **Attributes** provides the settings, which the object offers.

The settings are listed in the table of contents to the left.

The shared properties are described under the [Tab Attributes](#).

Services [Exporter]

To open a list, into which you type the names of the services, which the *Exporter* exports, click this button.



Remarks

Before you can type in data, click the **Inheritance** check box so that it looks like this .

- Type the name of the service into the cell, and click **Apply**. The *Exporter* attempts to export the services in the order in which you type them into the list.

Note

The name is not case-sensitive, just like the names of *attributes* and *methods* of the objects are not case-sensitive.

To save memory and improve access speed, all places which are using such a case-insensitive string are pointing to the same string in main memory. The visible and unexpected result is that the first occurrence of the string defines how the string is written in terms of upper- and lower-casing.

In *SimTalk* you can compare strings in an case-insensitive manner with the `~=` operator, compare [Relational Operators](#).

Plant Simulation inserts these services into the list of offered services of the *Broker*: To view them, select **View > Offered Services**.

You can use the commands on the [Context Menu of Embedded Lists](#) to edit the contents of the cells.

SimTalk

Services [SimTalk] - Exporter

hasService [SimTalk]

See also

Context Menu of Embedded Lists

Offered Services [Broker]

Fail Services [text box]

To interrupt the respective process for the duration of the failure if the *Exporter* provides services for an *importer*, select this check box.

☐ Fail Services ☒

Remarks

If *Plant Simulation* interrupts processing, for example because of a failure of the *Exporter*, *Plant Simulation* deletes the respective **Out** event and adds it anew at the end of the failure according to the remaining processing time.

To not interrupt the respective process of the *importer* at the beginning of a failure if the *Exporter* provides services for an *importer*, clear **Fail Services**. The respective event thus persists. During a failure new requests to the *Exporter* remain unsatisfied.

SimTalk

FailServices [SimTalk]

See also

Sub-tab Processing > [Release All Services When One Of Them Is Failed \[set-up importer\]](#)

Sub-tab Failure > [Release All Services When One Of Them Is Failed \[failure importer\]](#)

Priority [Exporter]

Type the **Priority** into the text box with which the *Exporter* carries out a work order. The **Priority** is a criterion for the urgency of a request.

Priority 

Remarks

The *Priority* is an integer value. The higher its value, the higher the urgency. *Plant Simulation* provides an *Exporter* with *Priority* 10 before providing an *Exporter* with *Priority* 1.

Exporters with a higher priority will be brokered earlier by the *Broker* for identical services than *Exporters* with a lower priority. The **Priority** only applies to *Exporters* who are registered with the same *Broker*.

If several qualified *Exporters* with the same priority exist, who can provide the requested service, *Plant Simulation* first brokers the *Exporters*, who already stay on a suitable *Workplace* at the station. After that *Exporters* are brokered who already are at the station, but stay on the wrong *Workplace*. Then *Exporters* are brokered who are not staying on a *Workplace*. Finally, those *Exporters* are brokered who stay on a *Workplace* that is assigned to another station.

SimTalk

Priority [SimTalk] - Exporter

Capacity [text box] - Exporter

Type the **Capacity** of the *Exporter* into the text box. The **Capacity** is a value greater than or equal to zero and designates the maximum number of services it can export.

Capacity 

Remarks

You can only reduce the capacity if the capacity provided to the *importers* is not fallen short of. If you increase the capacity, the *Broker* will immediately attempt to find new *importers*.

SimTalk

Capacity [SimTalk] - Exporter

Broker [Exporter]

Select the *Broker*, who assigns the services to the *Exporter*.

Broker: ... 

Remarks

Click the ellipsis button and select the *Broker*, who assigns the services to the *Exporter*, in the dialog **Select Object**.

Or type in the path to and the name of the *Broker*. You might, for example, type in `.Models.Model.MyBroker`.

Or select the *Broker* in a *Frame*, drag it to the text box and drop it there.

Note

F2 opens the dialog of the object, whose name you typed into the text box.

Plant Simulation enters the *Exporter* into the list of administered *Exporters* which this *Broker* manages.

To view them, select **View > Exporters**.

SimTalk

BrokerPath [SimTalk] - Exporter

See also

Broker [object]

Select Object

Tab Failures

Define **failures** as described under the [Tab Failures](#).

Tab Controls

Tab Controls [Exporter]

Click the ellipsis button and select a *Method* in the dialog **Select Object**. Type in the source code of the respective control.

Select the Path to an Existing Method

Click the ellipsis button. Navigate to the location of the *Method* in the dialog [Select Object \[for controls\]](#) and click **OK**. This inserts the name of the *Method* into the text box of the **Control**.



Press **F2** in the text box to open the *Method*. Then type in the source code of the **Control**.

Instead of choosing **Select Object**, you can also select the *Method* in a *Frame*, drag it to the text box and drop it there.

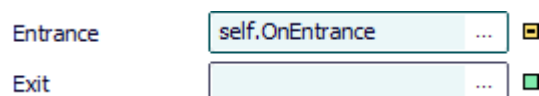
Create a Control That is a Method of the Object

Proceed as follows to create a control as a *user-defined attribute* of data type *Method*:

- Type a meaningful name into the text box and select [Create Control \[context menu\]](#). *Plant Simulation* then inserts `self.Name_you_typed_in_for_the_control`, such as `self.A1Ctrl`.



- Select **Create Control** on the empty text box. *Plant Simulation* then inserts `self.OnBuilt_in_name_of_the_control`, such as `self.OnEntrance`.



Type the source code of this control into the *Method* that opens.

To edit the source code later on:

- Press **F2**.
- Or hold down **Shift** and double-click into the text box.
- Or select **Open Object** on the context menu.
- Or click the tab **User-defined** and double-click the name of the *Method* in the list.
- To delete this control, delete the *user-defined attribute*. If you only delete the name from the text box, the *user-defined attribute* is retained.

See also

[Order Control \[Exporter\]](#)

[Release Control \[Exporter\]](#)

Order Control [Exporter]

Modifies the built-in behavior of the object. The **Order Control** specifies how the *Exporter* handles the order. The object calls the **Order Control** whenever the *Exporter* is assigned to an *importer*.



Select the Path to an Existing Method

Click the ellipsis button. Navigate to the location of the *Method* in the dialog [Select Object \[for controls\]](#) and click **OK**. This inserts the name of the *Method* into the text box of the **Control**.



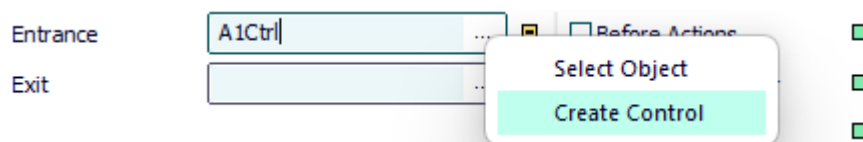
Press **F2** in the text box to open the *Method*. Then type in the source code of the **Control**.

Instead of choosing **Select Object**, you can also select the *Method* in a *Frame*, drag it to the text box and drop it there.

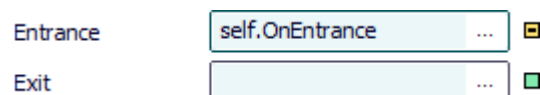
Create a Control That is a Method of the Object

Proceed as follows to create a control as a *user-defined attribute* of data type *Method*:

- Type a meaningful name into the text box and select **Create Control [context menu]**. *Plant Simulation* then inserts `self.Name_you_typed_in_for_the_control`, such as `self.A1Ctrl`.



- Select **Create Control** on the empty text box. *Plant Simulation* then inserts `self.OnBuilt_in_name_of_the_control`, such as `self.OnEntrance`.

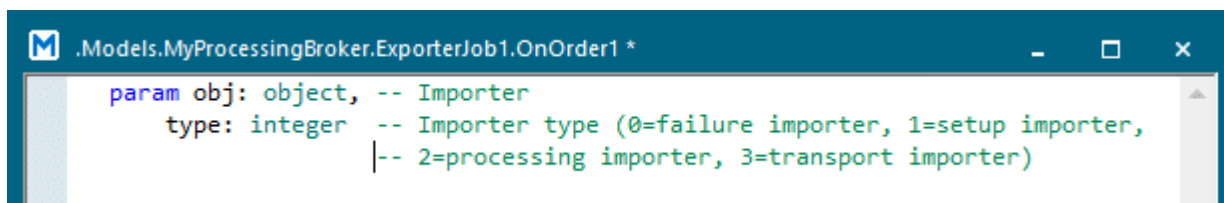


Type the source code of this control into the *Method* that opens.

To edit the source code later on:

- Press **F2**.
- Or hold down **Shift** and double-click into the text box.
- Or select **Open Object** on the context menu.
- Or click the tab **User-defined** and double-click the name of the *Method* in the list.
- To delete this control, delete the *user-defined attribute*. If you only delete the name from the text box, the *user-defined attribute* is retained.

The **standard order control** as a user-defined attribute looks like this:



Parameters

The **Order Control** has two parameters, which define the *importer*:

- The parameter *Importer* of data type *object* designates the *importer*.
- The parameter *Type* of data type *integer* designates its type: 0 designates the *failure/remove failure-importer*, 1 the *set-up-importer*, 2 the *processing-importer*, and 3 the *transport-importer*.

SimTalk

OrderCtrl [SimTalk] - Exporter

See also

Select Object [for controls]

Release Control [Exporter]

Modifies the built-in behavior of the object. The object calls the **Release Control** as soon as an *importer* releases the *Exporter/Worker*.

Remarks

Plant Simulation executes the **Release Control** as soon as an *importer* releases the *Exporter/Worker*. When the control is called, the *Exporter/Worker* has already left its *importer*. You yourself have to make sure that the *Exporter/Worker* is assigned new *importers*, for example with the method [findNewImporter](#). In this case the *Exporter/Worker* will not automatically register as being available with its *Broker*.

Select the Path to an Existing Method

Click the ellipsis button. Navigate to the location of the *Method* in the dialog [Select Object \[for controls\]](#) and click **OK**. This inserts the name of the *Method* into the text box of the **Control**.

Press **F2** in the text box to open the *Method*. Then type in the source code of the **Control**.

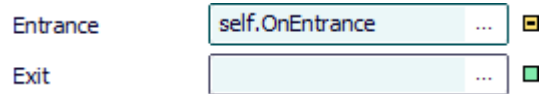
Instead of choosing **Select Object**, you can also select the *Method* in a *Frame*, drag it to the text box and drop it there.

Create a Control That is a Method of the Object

Proceed as follows to create a control as a *user-defined attribute* of data type *Method*:

- Type a meaningful name into the text box and select [Create Control \[context menu\]](#). *Plant Simulation* then inserts `self.Name_you_typed_in_for_the_control`, such as `self.A1Ctrl`.

- Select **Create Control** on the empty text box. *Plant Simulation* then inserts `self.OnBuilt_in_name_of_the_control`, such as `self.OnEntrance`.

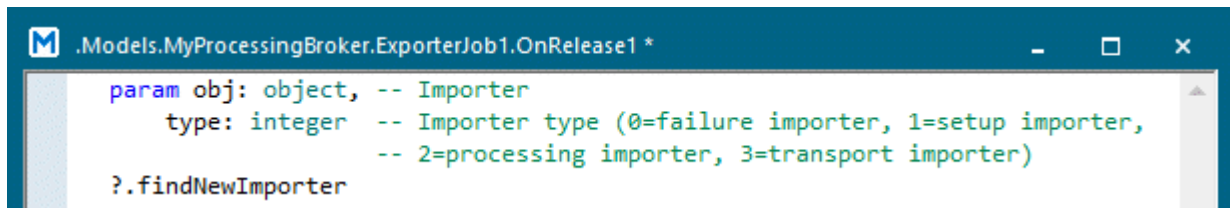


Type the source code of this control into the *Method* that opens.

To edit the source code later on:

- Press **F2**.
- Or hold down **Shift** and double-click into the text box.
- Or select **Open Object** on the context menu.
- Or click the tab **User-defined** and double-click the name of the *Method* in the list.
- To delete this control, delete the *user-defined attribute*. If you only delete the name from the text box, the *user-defined attribute* is retained.

The **standard release control** as a *user-defined attribute* looks like this:



Parameters

The **Release Control** has two parameters, which define the *importer*:

- The parameter *Importer* of data type *object* designates the *importer* proper.
- The parameter *Type* of data type *integer* designates its type:
0 designates the *failure/remove failure-importer*, 1 the *set-up-importer*, 2 the *processing-importer*, and 3 the *transport-importer*.

SimTalk

ReleaseCtrl [SimTalk] - Exporter

See also

[Select Object \[for controls\]](#)

Shift Calendar [Exporter]

Select the *ShiftCalendar*. It contains the data of the shifts in your installation and controls during which shifts the *Exporter* works in the plant.

Shift Calendar 

Remarks

Click the ellipsis button and select the *ShiftCalendar* in the dialog **Select Object**.

Instead of clicking the ellipsis button, you can also select the *ShiftCalendar* in a *Frame*, drag it to the text box, and drop it there.

This automatically enters the object into the list of **Objects** on the tab **Resources** of the *ShiftCalendar*.

SimTalk

[ShiftCalendarObject \[SimTalk\] - material flow objects](#)

See also

[ShiftCalendar \[object\]](#)

[Select Object \[for controls\]](#)

Tab Statistics

Tab Statistics [Exporter]

The tab **Statistics** shows the most important statistical data.

Remarks

The tab shows the values for the *services* and for the *Exporter*. For each of these blocks, adding up the individual values results in a value 100 percent.

Note

To show the statistics values for the *services* in a *Chart*, drag the *Exporter* over a *Chart* and drop it there.
To show the statistics values for the *Services* and for the *Exporter* in a *Chart*, hold down **Shift**, drag the *Exporter* over a *Chart* and drop it there.

To collect the failure times of the *services* and of the *Exporter*, select the check box **Fail Services**.

To only collect the failure times of the *Exporter*, clear the check box **Fail Services**.

To collect *Exporter* statistics data, select the check box **Exporter Statistics** ☒ **Exporter Statistics** ☐.

The *Exporter* only collects *failed* times that accrue during the processing and setup time, i.e., times that are located outside of the *paused* and of the *unplanned* time.

Waiting times only accrue during the times when the *Exporter* is available, i.e., when these times are located outside of the *paused*, the *unplanned*, and the *failed* times.

Note

The *Exporter* stays at the station, to which he is brokered, while this station is **failed**, **paused**, or **stopped**. Statistics does not count this time as **waiting**.

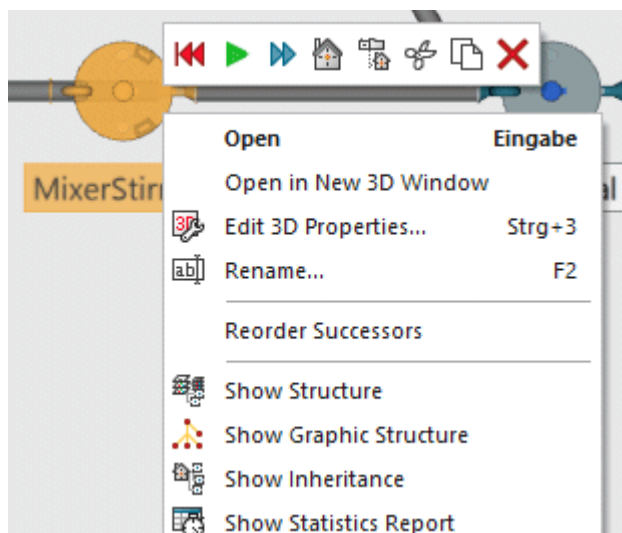
Attributes		Failures	Controls	Statistics	User-defined
☒ Exporter Statistics ☐					
Services	Working	81.25%		Free Capacity	0
	Setting-up	0.00%		Mediated Capacity	1
	Repairing	0.00%		Free Capacity (sum)	173158
	Waiting	18.75%		Mediated Capacity (sum)	173159
	Failed	0.00%		Minimum Free Capacity	0
Exporter	Operational	100.00%		Maximum Free Capacity	1
	Paused	0.00%		Minimum Med. Capacity	0
	Unplanned	0.00%		Maximum Med. Capacity	1
	Failed	0.00%			

Item English	Description	Read-only attribute	Item German
Services Processing	Shows the portion of the processing time of the services of the overall statistics time of the <i>Exporter</i> , weighted with the capacity. The overall statistics time is the statistics collection period without the unplanned time and the paused time .	StatServicesWorkingPortion [SimTalk] - Exporter	Dienste arbeitend
Services Setting-up	Shows the portion of set-up time of the services of the overall statistics time of the <i>Exporter</i> , weighted with the capacity. The overall statistics time is the statistics collection period without the unplanned time and the paused time .	StatServicesSetupPortion [SimTalk] - Exporter	Dienste rüstend
Services Repairing	Shows the portion of the repairing time of the overall statistics time of the <i>Exporter</i> , weighted with the capacity. The overall statistics time	StatServicesRepairingPortion [SimTalk] - Exporter	Dienste reparierend

Item English	Description	Read-only attribute	Item German
	is the statistics collection period without the unplanned time and the paused time .		
Services Waiting	Shows the portion of the waiting time of the services for an <i>importer</i> of the overall statistics time of the <i>Exporter</i> , plus the time the <i>services</i> spent waiting for a <i>part</i> at the <i>importer</i> , weighted with the capacity. The overall statistics time is the statistics collection period without the unplanned time and the paused time .	StatServicesWaitingPortion [SimTalk] - Exporter StatServicesWaitingImpPortion [SimTalk] - Exporter StatServicesWaitingMUPortion [SimTalk] - Exporter	Dienste wartend
Services Failed	Shows the portion of the failed time of the services of the overall statistics time of the <i>Exporter</i> , weighted with the capacity. The overall statistics time is the statistics collection period without the unplanned time and the paused time .	StatServicesFailedPortion [SimTalk] - Exporter	Dienste gestört
Exporter Operational	Shows the portion of the statistics collection period during which the <i>Exporter</i> was operational, weighted with the capacity. The overall statistics time is the statistics collection period without the unplanned time and the paused time .	StatExporterOperationalPortion [SimTalk]	Exporter einsatzbereit
Exporter Paused	Shows the portion of the statistics collection period during which the <i>Exporter</i> was paused.	StatExporterPausedPortion [SimTalk]	Exporter pausiert
Exporter Unplanned	Shows the portion of the statistics collection period during which the <i>Exporter</i> was unplanned.	StatExporterUnplannedPortion [SimTalk]	Exporter ungeplant
Exporter failed	Shows the portion of the statistics collection period during which the <i>Exporter</i> was failed.	StatExporterFailedPortion [SimTalk]	Exporter gestört
Free Capacity	Shows the capacity that is available at the moment.	FreeCapacity [SimTalk]	Freie Kapazität
Mediated Capacity	Shows the capacity that is brokered at the moment.	MediatedCapacity [SimTalk] - Exporter	Belegte Kapazität
Free Capacity (sum)	Shows the sum of the released capacity that the <i>Exporter</i> placed with any <i>importer</i> .	StatSumFreeCapacity [SimTalk]	Summe freie Kapazität

Item English	Description	Read-only attribute	Item German
Mediated capacity (sum)	Shows the sum of the capacities that are brokered at the moment	StatSumMediatedCapacity [SimTalk]	Summe belegte Kapazität
Minimum Free Capacity	Shows the minimum capacity that is available. If the <i>Exporter</i> did not collect any valid value yet, the tab shows -1.	StatMinFreeCapacity [SimTalk]	Minimum freie Kapazität
Maximum Free Capacity	Shows the maximum capacity that is available. If the <i>Exporter</i> did not collect any valid value yet, the tab shows -1.	StatMaxFreeCapacity [SimTalk]	Maximum freie Kapazität
Minimum Mediated Capacity	Shows the minimum occupied capacity. If the <i>Exporter</i> did not collect any valid value yet, the tab shows -1.	StatMinMediatedCapacity [SimTalk]	Minimum belegte Kapazität
Maximum Mediated Capacity	Shows the maximum occupied capacity. If the <i>Exporter</i> did not collect any valid value yet, the tab shows -1.	StatMaxMediatedCapacity [SimTalk]	Maximum belegte Kapazität

To view [Exporter and Worker Statistics](#) in the [Statistics Report](#), select **View > Show Statistics Report** in the dialog of the object. You can also click the right mouse button in the *Frame* and select [Show Statistics Report](#) or you can press **F6**.



Exporter Statistics [Exporter]

To collect statistics data of the *Exporter*, select this check box. To deactivate statistics collection, clear the check box.

☒ **Exporter Statistics** ☐

SimTalk

[ExpStatOn \[SimTalk\]](#)

Tab User-defined

Define your own attributes as described under the [Tab User-defined](#).

Navigate Menu

The commands are described under the [Navigate Menu](#).

View Menu

View Menu [Exporter]

The **View Menu** provides commands to access its functions.

Refresh [on View menu]	Importers [Exporter]
Show Statistics Report [on View menu]	Exported Services [Exporter]
Show Attributes and Methods [on View menu]	Associated Shift Calendar

See also

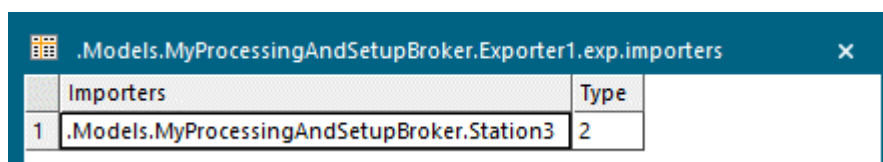
[View Menu \[general description\]](#)

Importers [Exporter]

Opens a table, which shows all *importers* for which the *Exporter* provides services.

Remarks

The table shows the path to and the names of the **Importers** and their **Type**: The **Type** 0 designates the *failure-importer*, 1 the *set-up-importer*, 2 the *processing-importer*, and 3 the *transport-importer*.



	Importers	Type
1	.Models.MyProcessingAndSetupBroker.Station3	2

See also

[findNewImporter \[SimTalk\]](#)

[getImporters \[SimTalk\]](#)

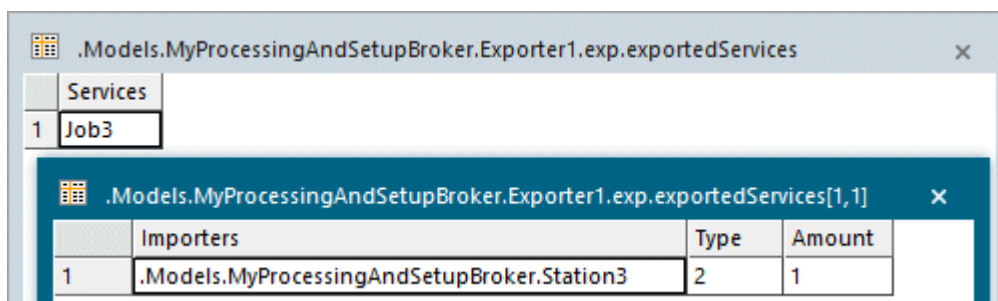
Exported Services [Exporter]

Opens a table, which shows the names of all services which the *Exporter* exports at this time.

Remarks

Double-click the name of a service to open a subtable.

The subtable shows the name of the **Importer**, its **Type** (0 designates the *failure-importer*, 1 the *set-up-importer*, 2 the *processing-importer*, and 3 the *transport-importer*) and the **Amount** of services provided.

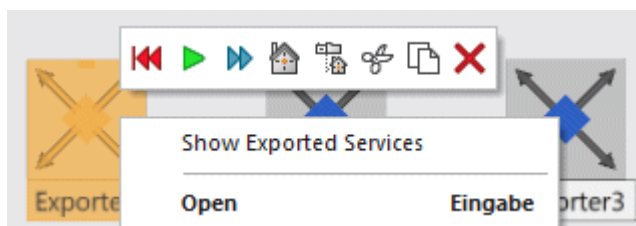


The screenshot shows two overlapping table windows. The top window, titled '.Models.MyProcessingAndSetupBroker.Exporter1.exp.exportedServices', contains a table with one row: '1 Job3'. The bottom window, titled '.Models.MyProcessingAndSetupBroker.Exporter1.exp.exportedServices[1,1]', contains a table with three columns: 'Importers', 'Type', and 'Amount'. It has one row: '1 .Models.MyProcessingAndSetupBroker.Station3', '2', and '1'.

.Models.MyProcessingAndSetupBroker.Exporter1.exp.exportedServices	
1	Job3

.Models.MyProcessingAndSetupBroker.Exporter1.exp.exportedServices[1,1]		
Importers	Type	Amount
1 .Models.MyProcessingAndSetupBroker.Station3	2	1

Instead, you can also right-click the *Exporter* and select **Show Exported Services** on the context menu in the *Frame*.



See also

[getExportedServices \[SimTalk\]](#)

[Services \[SimTalk\] - Exporter](#)

Tools Menu

Tools Menu

The **Tools Menu** provides these menu commands:

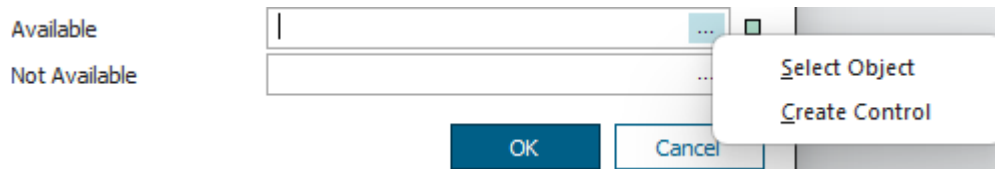
[Edit Controls \[material flow objects\] > Available Control \[Exporter\]](#)

[Edit Controls \[material flow objects\] > Not Available Control \[Exporter\]](#)

Edit Observers [general description]

Available Control [Exporter]

Modifies the built-in behavior of the object. The object calls the **Available Control** whenever the *Exporter/Worker* changes his state to **available**.



Select the Path to an Existing Method

Click the ellipsis button. Navigate to the location of the *Method* in the dialog [Select Object \[for controls\]](#) and click **OK**. This inserts the name of the *Method* into the text box of the **Control**.



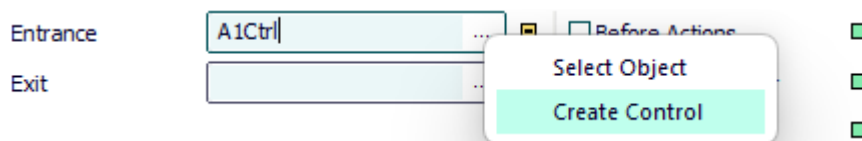
Press **F2** in the text box to open the *Method*. Then type in the source code of the **Control**.

Instead of choosing **Select Object**, you can also select the *Method* in a *Frame*, drag it to the text box and drop it there.

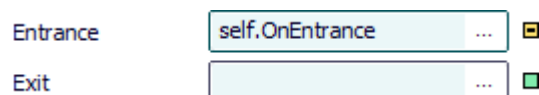
Create a Control That is a Method of the Object

Proceed as follows to create a control as a *user-defined attribute* of data type *Method*:

- Type a meaningful name into the text box and select **Create Control [context menu]**. *Plant Simulation* then inserts `self.Name_you_typed_in_for_the_control`, such as `self.A1Ctrl`.



- Select **Create Control** on the empty text box. *Plant Simulation* then inserts `self.OnBuilt_in_name_of_the_control`, such as `self.OnEntrance`.



Type the source code of this control into the *Method* that opens.

To edit the source code later on:

- Press **F2**.
- Or hold down **Shift** and double-click into the text box.
- Or select **Open Object** on the context menu.
- Or click the tab **User-defined** and double-click the name of the *Method* in the list.
- To delete this control, delete the *user-defined attribute*. If you only delete the name from the text box, the *user-defined attribute* is retained.

SimTalk

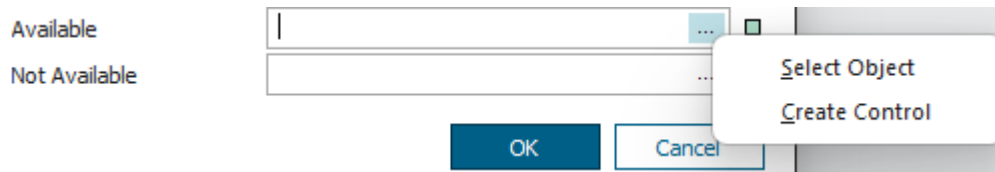
[AvailableCtrl \[SimTalk\]](#)

See also

[Select Object \[for controls\]](#)

Not Available Control [Exporter]

Modifies the built-in behavior of the object. The object calls the **Not Available Control** whenever the *Exporter/Worker* changes his state to **not available**.



Select the Path to an Existing Method

Click the ellipsis button. Navigate to the location of the *Method* in the dialog [Select Object \[for controls\]](#) and click **OK**. This inserts the name of the *Method* into the text box of the **Control**.



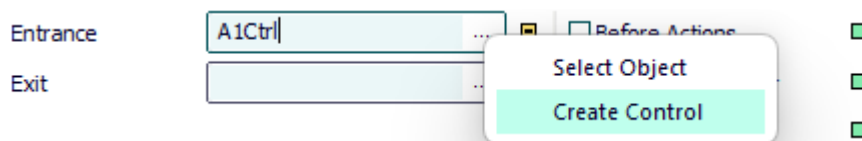
Press **F2** in the text box to open the *Method*. Then type in the source code of the **Control**.

Instead of choosing **Select Object**, you can also select the *Method* in a *Frame*, drag it to the text box and drop it there.

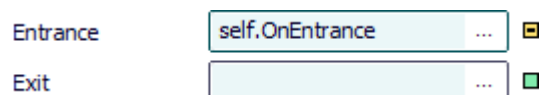
Create a Control That is a Method of the Object

Proceed as follows to create a control as a *user-defined attribute* of data type *Method*:

- Type a meaningful name into the text box and select **Create Control [context menu]**. *Plant Simulation* then inserts `self.Name_you_typed_in_for_the_control`, such as `self.A1Ctrl`.



- Select **Create Control** on the empty text box. *Plant Simulation* then inserts `self.OnBuilt_in_name_of_the_control`, such as `self.OnEntrance`.



Type the source code of this control into the *Method* that opens.

To edit the source code later on:

- Press **F2**.
- Or hold down **Shift** and double-click into the text box.
- Or select **Open Object** on the context menu.
- Or click the tab **User-defined** and double-click the name of the *Method* in the list.
- To delete this control, delete the *user-defined attribute*. If you only delete the name from the text box, the *user-defined attribute* is retained.

SimTalk

[NotAvailableCtrl \[SimTalk\]](#)

See also

[Select Object \[for controls\]](#)

Help Menu

The commands are described under the [Help Menu](#).

States of the Exporter

During the simulation runs *Plant Simulation* shows the **state of the Exporter** as one or more **horizontally arranged cubes** along the top of the picture of the object in the *Frame*.

Remarks

These states are designated by different colors.

Description	Color of the State Graphic
The <i>Exporter</i> is failed .	red
The <i>Exporter</i> exports services.	green
The <i>Exporter</i> is paused .	blue

See also

[Tab States](#) > [State Graphic \[defined\]](#) in 3D

[States of the Material Flow Objects](#)

[States of the Energy Features](#)

[States of the Frame](#)

[States of the Transporter](#)

[States of the Method](#)

[States of the MQTT Interface](#)

Methods of the Exporter

_Methods of the Exporter

The *Exporter* provides:

- The *methods* listed in the table of contents to the left.
- The [Methods of All Objects](#).